

Recap about static and dynamical systems, written in yesterday's notes.

Today we focus on the solution of linear continuous time state space model.

Here is the model:

$$\begin{cases} \dot{x} = Ax + Bu \\ y = Cx + Du \end{cases}$$

So, we want to find the solution of the model:

$$\begin{cases} x(t) = ? \\ y(t) = ? \end{cases}$$

We recognise dynamical systems when we have a derivative among variables.

Derivatives don't tell us about the absolute value of a variable, rather the rate of change of a variable.

Remember that we must know $x(t_0)$ to find the solution.

Here is the formula for the solution of state space model (linear continuous time).

For the setting, we have a dynamical system. t_0 is the initial time, $x(t_0)$ is the initial value of the state variables. We apply input $u_{[t_0, t]}$.

Our task is to compute $y(t)$ and $x(t)$ for $t \geq t_0$.

Yesterday we've seen this:

$$\begin{cases} x(t) = \Phi(t_0, t, x(t_0, u_{[t_0, t]})) \\ y(t) = \gamma(t_0, t, x_0, u_{[t_0, t]}) \end{cases} \quad \Phi \text{ is the state transition function}$$

Here is the abstract, high-level formula of the solution:

$$x(t) = e^{A(t-t_0)}x(t_0) + \int_{t_0}^t e^{A(t-\tau)} \cdot B \cdot u(\tau) d\tau; x(t_0) \text{ known}$$

$$y(t) = C \cdot e^{A(t-t_0)}x(t_0) + \int_{t_0}^t C \cdot e^{A(t-\tau)} \cdot B \cdot u(\tau) d\tau + Du(t)$$

$e^{A(t-t_0)}$ is the exponential of a matrix. What is it?

It has the same definition of a standard exponential of a number.

$$e^{A(t-t_0)} = \sum_{k=0}^{\infty} \frac{1}{k!} \cdot A^k \cdot t^k$$

The problem is, we cannot use this formula, as it has ∞ many terms, and computing A^k is not obvious. In fact, we have that $A^0 = I$, $A^1 = A$, but it becomes complicated to compute $A^2, \dots$

Today, we're going to see in practice how to compute the formulas above.

Taking a look at x , we see it's composed of two terms. the first one is the free evolution of system in the state, while the second is the forced state evolution. Similarly, for the output. The first terms is the free output evolution, the second one is the forced output evolution (including $Du(t)$ as well).

Let's pick the example of a mass linked to a wall with a spring and dumper, and it's subject to a force.

The state and output of the system moves for two reasons: first as we're giving an input (with a force), but also because of the initial state of the system, which is not necessarily at rest. Imagine the spring was extended in the initial state, then the system would move due to characteristics of a spring.

Now, how do we use in practice the formula? We know a system (A, B, C, D) , and we want to compute $x(t), y(t)$.

The general difficult case is this one: a system with many inputs, outputs and state variables.

Let's focus on the free evolution of the state first, $x_{free}(t) = e^{A(t-t_0)} \cdot x(t_0)$.

The simplest case is a system with one state $x(t) = x_1(t)$. It's actually trivial, as the matrix exponential is the standard exponential of a number.

By having $\begin{cases} \dot{x} = a \cdot x \\ x(t_0) \text{ known} \end{cases}$ we have $x(t) = e^{a(t-t_0)}x(t_0)$, obvious case.

Let's work now on case two. We have now many state variables (n_x).

We have the generic system

$$\begin{cases} \dot{x} = A \cdot x \\ x(t_0) \text{ known} \end{cases} \leftrightarrow \begin{cases} \dot{x}_1 = a_{11}x_1 + a_{12}x_2 + \dots + a_{1n_x}x_{n_x} \\ \dot{x}_2 = a_{21}x_1 + a_{22}x_2 + \dots + a_{2n_x}x_{n_x} \\ \vdots \\ \dot{x}_n = a_{n1}x_1 + a_{n2}x_2 + \dots + a_{nn_x}x_{n_x} \end{cases}$$

By keeping only the diagonal coefficients equal to one, and the other equal to zero, we're replicating the simple case.

So, for case 2, we have a set of n_x decoupled system with one state variable.

Let's define diagonal coefficients as λ .

$$\begin{cases} \dot{x}_1 = \lambda_1 x_1 \\ \dot{x}_2 = \lambda_2 x_2 \\ \vdots \\ \dot{x}_n = \lambda_n x_n \end{cases}$$

With decouple we mean that sub-systems do not influence each other. The solution is now trivial, as we simply have to solve each equation independently.

$$\begin{cases} x_1(t) = e^{\lambda_1(t-t_0)} x_1(t_0) \\ x_2(t) = e^{\lambda_2(t-t_0)} x_2(t_0) \\ \vdots \\ x_{n_x}(t) = e^{\lambda_{n_x}(t-t_0)} x_{n_x}(t_0) \end{cases}$$

What is the matrix form?

$$x(t) = \begin{pmatrix} x_1 \\ x_2 \\ \dots \\ x_n \end{pmatrix} = \underbrace{\begin{pmatrix} e^{\lambda_1(t-t_0)} & 0 & \dots & 0 \\ 0 & e^{\lambda_2(t-t_0)} & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & e^{\lambda_n(t-t_0)} \end{pmatrix}} \cdot \begin{pmatrix} x_1(t_0) \\ x_2(t_0) \\ \dots \\ x_n(t_0) \end{pmatrix}$$

So, recap for case 2. We started from the independent equations $\dot{x}_1 = \lambda_1 x_1$, etc.

This is scalar, and equivalent to a diagonal matrix of λ s multiplied by x .

This matrix is called as Λ (capital λ) matrix.

We then solve all the decoupled equations in the form $x_1 = e^{\lambda_1(t-t_0)} x_1(t_0)$, etc.

We can write this in matrix form, getting

$$x = \begin{pmatrix} e^{\lambda_1(t-t_0)} & 0 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & \dots & \dots & e^{\lambda_n(t-t_0)} \end{pmatrix} x(t_0)$$

The matrix is called $e^{\Lambda(t-t_0)}$.

When A matrix is diagonal (called Λ to stress this fact), it is very simple to compute its matrix exponential $e^{\Lambda(t-t_0)}$. In fact, it is simply a diagonal matrix, with the generic element on the diagonal, $(e^{\Lambda(t-t_0)})_{ii}$, equal to the time exponential of $\Lambda_{ii} = \lambda_i$, so:

$$(e^{\Lambda(t-t_0)})_{ii} = e^{\lambda_i(t-t_0)}$$

Now, let's take a look at case 3. It's the case in which A is non diagonal, so a generic A .

From case 2, we know that solution is easy to compute when A is diagonal.

Given a system, by writing the physical laws that govern it, we find the (natural) model

$$\begin{cases} \dot{x} = Ax + Bu \\ y = Cx + Du \end{cases}$$

We saw that we can define a change of coordinates to find new models.

For a system, we can have infinite many equivalent models.

Having a model in x , and going to an equivalent model z , we have a model like this.

$$\begin{cases} \dot{z} = \underbrace{T^{-1}AT}_{\tilde{A}}z + \underbrace{T^{-1}B}_{\tilde{B}} \\ y = \underbrace{TC}_{\tilde{C}}z + Du \end{cases}$$

Having $x(t_0)$, we have that $z(t_0) = T^{-1}x(t_0)$

The question now is: can I find a change of coordinates $z = T^{-1}x$ such that $\tilde{A} = (T^{-1}AT) = \Lambda$ (is diagonal)?

The scheme is this:

$$\dot{x} = Ax \xrightarrow{z=T^{-1}x} \dot{z} = T^{-1}ATz = \Lambda z$$

The solution for this is $z(t) = e^{\Lambda(t-t_0)} \cdot z(t_0)$, and it's case two on z coordinates.

From this, we need to go back, with $x = Tz$, to $x(t) = ?$.

We have that $z(t) = e^{\Lambda(t-t_0)} \cdot z(t_0)$ in x variables is given by

$$T^{-1}x(t) = e^{\Lambda(t-t_0)}T^{-1}x(t_0).$$

Multiplying both sides on the left by T , we get the new formula:

$$x(t) = T \cdot e^{\Lambda(t-t_0)}T^{-1}x(t_0).$$

So, the final formula for $x(t)$, substituting Λ with $T^{-1}AT$, is:

$$x(t) = T \cdot e^{T^{-1}AT(t-t_0)}T^{-1}x(t_0)$$

We have that $e^{T^{-1}AT(t-t_0)}$ is simple to compute, and the rest is also doable.

While, what is $T \cdot e^{T^{-1}AT(t-t_0)}T^{-1}$? It's the matrix exponential $e^{A(t-t_0)}$, computed in a doable way.

So, for case 3: if T^{-1} exists such that $T^{-1}AT = \Lambda$ then $e^{A(t-t_0)} = T \cdot e^{\Lambda(t-t_0)}T^{-1}$, and easy to compute. This is the spectral formula of matrix exponential.

Then, we have case 4: how to find T^{-1} such that $T^{-1}AT = \Lambda$ diagonal?

We can compact the diagonal matrix Λ as $diag(\lambda_1, \lambda_2, \dots, \lambda_n)$.

The key idea is to assume we have solved the problem and so $T^{-1}AT$ is diagonal. We don't know at the moment how to find T , but let's assume we solved it. Let's find interesting properties that such T^{-1} satisfies. These are the necessary conditions that a T^{-1} matrix that diagonalises the system must respect.

Our analysis starts from $T^{-1}AT = \Lambda$. From this one, we want to learn interesting properties for matrix T . Let's look for matrices satisfying these properties.

T^{-1} is such that $T^{-1}AT = \Lambda$. Let's study this equation. A is a $n \times n$ matrix, with $n = n_x$.

Λ is a diagonal matrix $\text{diag}(\lambda_1, \lambda_2, \dots, \lambda_n)$, being a $n \times n$ matrix.

T^{-1} is a $n \times n$ matrix, to be found.

T is the inverse of T^{-1} , such that $TT^{-1} = T^{-1}T = I$.

We can identify the columns for matrix T , which we call as u_i .

$$T = \begin{pmatrix} | & | & \dots & | \\ u_1 & u_2 & \dots & u_n \\ | & | & \dots & | \end{pmatrix}_{n \times n}$$

Multiply the equation $T^{-1}AT = \Lambda$ on the left by T , obtaining $TT^{-1}AT = T\Lambda$, thus remaining with $AT = T\Lambda$.

We rewrite T highlighting the columns

$$A[u_1, u_2, \dots, u_n] = [u_1, u_2, \dots, u_n]\Lambda$$

How do we rewrite this equation now?

$$(A)_{n \times n} \cdot \begin{pmatrix} | & | & \dots & | \\ u_1 & u_2 & \dots & u_n \\ | & | & \dots & | \end{pmatrix} = \begin{pmatrix} | & | & \dots & | \\ u_1 & u_2 & \dots & u_n \\ | & | & \dots & | \end{pmatrix} \begin{pmatrix} \lambda_1 & & & \\ 0 & \lambda_2 & & \\ \dots & & \dots & \\ 0 & 0 & \dots & \lambda_n \end{pmatrix}$$

On the first and second member we'll have a new $n \times n$ matrix, resulting in:

$$\begin{pmatrix} | & | & \dots & | \\ c_1 & c_2 & \dots & c_n \\ | & | & \dots & | \end{pmatrix} = \begin{pmatrix} | & | & \dots & | \\ c'_1 & c'_2 & \dots & c'_n \\ | & | & \dots & | \end{pmatrix}$$

Equivalent to:

$$[Au_1, Au_2, \dots, Au_n]_{n \times n} = [\lambda_1 u_1, \lambda_2 u_2, \dots, \lambda_n u_n]$$

The columns of T satisfies these important properties:

$$Au_i = \lambda_i u_i \quad \text{for generic } i = 1, 2, \dots, n$$

Remember that A is a $n \times x$ matrix, u_i is column vector of n numbers, λ_i is a number.

The equation above tells us that u_i is an eigenvector of A , and λ_i is the associated eigenvalue.

Definition of eigenvectors and eigenvalues:

Given an $n \times n$ matrix A , a column vector $u \neq 0$ is called an eigenvector of A if it exists one number $\lambda \in \mathbb{C}$ such that $Au = \lambda u$. λ is called the associated eigenvalue.

Let's take an example:

$A = \begin{pmatrix} 1 & 0 \\ 1 & 2 \end{pmatrix}$. By picking a vector $v = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$, we get a new vector from their product: $A \cdot v = \begin{pmatrix} 1 \\ 5 \end{pmatrix}$. It's totally different from $\lambda \begin{pmatrix} 1 \\ 2 \end{pmatrix}$.

With eigenvectors we remain in the same direction of a vector, but we change the amplitude of it.

Conclusion: if T^{-1} is such that $(T^{-1}AT) = \Lambda$, then:

- $Au_i = \lambda_i u_i$, where u_i is the i -th column of T , and λ_i is the i -th diagonal element of T^{-1} .
1. u_i (the columns of T) are eigenvectors of A .
 2. λ_i are the associated eigenvalues.

To find a suitable transformation T , we must be able to find the eigenvectors and eigenvalues of matrix A .

If T is such that $T^{-1}AT = \Lambda$, $T = [u_1, u_2, \dots, u_n]$ are eigenvectors of A . But also the opposite is true.

By taking a matrix $T = [u_1, u_2, \dots, u_n]$ composed by putting together eigenvectors of A , then $T^{-1}AT = \Lambda$, or $AT = T\Lambda$.